k^n $\quad e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \dots e_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$ $\quad$ base ?

$v \in k^n \quad v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \dots + x_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$

$= x_1 e_1 + \dots + x_n e_n$

$v \in \mathrm{Sp}\{e_1 \dots e_n\}$

$\mathrm{Sp}\{e_1 \dots e_n\} = k^n$

$\lambda_1 \dots \lambda_n \in k \quad \underbrace{\lambda_1 e_1 + \dots + \lambda_n e_n}_{\begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}} = 0_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

$\Rightarrow \lambda_1 = \dots = \lambda_n = 0 \Rightarrow e_1 \dots e_n$ lin. indep.

$\Rightarrow \{e_1 \dots e_n\}$ base di k^n

$k[x]$ $\quad B = \{1, x, x^2, \dots, x^n, \dots\}$

baza

lin. indep. $\quad \{x^{n_1}, x^{n_2}, \dots, x^{n_p}\}$

$n_1 < n_2 < \dots < n_p$!

Fie $\lambda_1 \dots \lambda_p \in k$ a.i

$\lambda_1 x^{n_1} + \dots + \lambda_p x^{n_p} = 0$

$\Rightarrow \lambda_1 = \dots = \lambda_p = 0$

$2x^3, \; -x^3, \; -x^3$

Sist. de gen. $\quad P(x) = a_0 + \dots + a_n x^n$

$n = \text{grad } P$

$\Rightarrow P(x) = a_0 \cdot 1 + a_1 \cdot x + \dots + a_n \cdot x^n$

$\operatorname{Sp}\{1, x, x^2, \dots\} = k[x]$

$V = M_2(\mathbb{R})$ $\quad k = \mathbb{R}$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{E_{11}} + b\underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{E_{12}} + c\underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{E_{21}} + d\underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{E_{22}}$$

$4 = 2 \cdot 2$

$\mathrm{Sp}\{E_{11}, E_{12}, E_{21}, E_{22}\} = M_2(\mathbb{R})$

$(E_{ij})_{i,j \in \overline{1,2}}$ lin. indep $\Big\}$ bază

$k[X]$ $\quad \dim_k k[X] = \infty$

$k_n[X] = \{P \in k[X] \mid \mathrm{grad}\, P \leq n\}$

$\{1, X, \dots, X^n\}$

$\dim_k k_n[X] = n+1$!

Prop. k corp V k-Sp. vect

$\mathcal{B} = \{v_1 \dots v_n\}$ bază

$w_1 \dots w_m \in V$ lin. indep. $\Rightarrow m \leq n$

Dem.: Pp. că $m > n$

$\forall i = \overline{1,m}$ $\exists \alpha_{ij} \in k$ $\underset{j=\overline{1,n}}{}$ a.î. $w_i = \sum_{j=1}^{n} \alpha_{ij} v_j$

Fie $\lambda_1 \dots \lambda_m \in k$ a.î. $\sum_{i=1}^{m} \lambda_i w_i = 0$

$$0 = \sum_{i=1}^{m} \lambda_i w_i = \sum_{i=1}^{m} \lambda_i \sum_{j=1}^{n} \alpha_{ij} v_j =$$

$$= \sum_{j=1}^{n} \underbrace{\left(\sum_{i=1}^{m} \lambda_i \alpha_{ij} \right)}_{\in k} v_j \Rightarrow$$

$v_1 \dots v_n$ lin. indep.

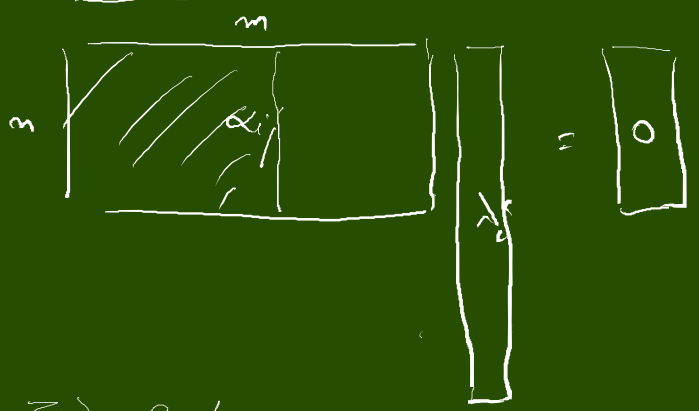
$$\Rightarrow \sum_{i=\overline{1,m}} \lambda_i a_{ij} = 0 \qquad \forall j = \overline{1,n}$$

Sistem liniar omogen de m ec. $(\lambda_i)_{i=\overline{1,m}}$

n ec.

m nec.

$m > n$



$\Rightarrow$ Sist compatibil nedeterminat

$\exists \lambda_i \neq 0$ sol. pt. Sist.

$\Rightarrow$ contrad.

$v_1 \dots v_n$ base $\iff \forall v \in V \; \exists! \lambda_1 \dots \lambda_n \in k$

a.i. $v = \lambda_1 v_1 + \dots + \lambda_n v_n$

lin. indep. syst. de gen.

$\Rightarrow$ " Se $v \in V$ pp. $\exists \lambda_1 \dots \lambda_n \in k$

$v = \lambda_1 v_1 + \dots + \lambda_n v_n$

" $\Leftarrow$ " tema

$\exists \mu_1 \dots \mu_n \in k$

$v = \mu_1 v_1 + \dots + \mu_n v_n$

$\Rightarrow \quad 0 = (\lambda_1 - \mu_1) v_1 + \dots + (\lambda_n - \mu_n) v_n$

$v_1 \dots v_n$ lin. indep $\Rightarrow$

$\Rightarrow \lambda_1 = \mu_1 \; \dots \; \lambda_n = \mu_n$

Exemple. $V = \mathbb{R}^2$ $k = \mathbb{R}$ $\dim_{\mathbb{R}} \mathbb{R}^2 = 2$

a) $\left\{ \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ NU

b) $\left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$ NU

c) $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ DA

d) $\left\{ \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} \right\}$?

c) $B = \left\{ v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

$\forall v \in \mathbb{R}^2$ $v = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \exists!\ \lambda_1, \lambda_2 \in \mathbb{R}$

$v = \lambda_1 v_1 + \lambda_2 v_2$

$$\begin{cases} x = \lambda_1 + \lambda_2 \\ y = 2\lambda_1 + \lambda_2 \end{cases} \qquad \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = (v_1 | v_2)$$

$\Rightarrow \lambda_1 = y - x, \quad \lambda_2 = 2x - y$

$\Rightarrow [v]_B = \begin{pmatrix} y - x \\ 2x - y \end{pmatrix}$

Corolar: $v_1 \dots v_n \in k^n$ baza $\Leftrightarrow$

$\Leftrightarrow \det(v_1 | \dots | v_n) \neq 0$

$\mathcal{B} = \{v_1, \dots, v_n\}$

$\forall v \in V \quad \exists! \lambda_1 \dots \lambda_n \in k$

$v = \lambda_1 v_1 + \dots + \lambda_n v_n$

$v_1 = 1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n$

$[v_1]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad [v_2]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = e_2 \dots$

e_1

$V_1 = \mathbb{R}^4$ $\quad$ $V_2 = \mathcal{M}_2(\mathbb{R})$ (2·2) $\quad$ $V_3 = \mathbb{R}_3[X]$ (3+1)

$\mathcal{B}_1 = \{e_1 \dots e_4\}$ $\quad$ $\mathcal{B}_2 = \{E_{ij}\}$ $\quad$ $\mathcal{B}_3 = \{1, X, X^2, X^3\}$

$$v = \begin{pmatrix} 1 \\ -3 \\ 0 \\ 2 \end{pmatrix} \qquad [v]_{B_1} = \begin{pmatrix} 1 \\ -3 \\ 0 \\ 2 \end{pmatrix} = v \qquad \mathbb{K}^4$$

$$v = 1 \cdot e_1 + (-3) e_2 + 0 \cdot e_3 + 2 \cdot e_4$$

$$A = \begin{pmatrix} 1 & -3 \\ 0 & 2 \end{pmatrix} = 1 \cdot E_{11} + (-3) E_{12} + 0 \cdot E_{21} + 2 \cdot E_{22}$$

$$[A]_{B_2} = \begin{pmatrix} 1 \\ -3 \\ 0 \\ 2 \end{pmatrix}$$

$$P(X) = 1 - 3X + 2X^3$$

$$= 1 \cdot 1 + (-3) X + 0 \cdot X^2 + 2 \cdot X^3$$

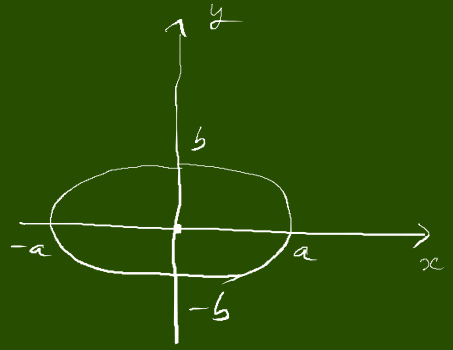
$$[P]_{B_3} = \begin{pmatrix} 1 \\ -3 \\ 0 \\ 2 \end{pmatrix}$$

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exemples :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\{(x,y) \,/\, \underbrace{18x^2 + 24xy + 11y^2 = 50}\}$$



$$\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\} \longrightarrow$$

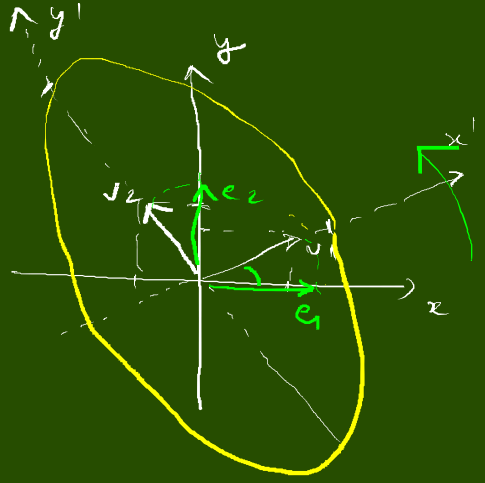
$$\longrightarrow \left\{\underbrace{\begin{pmatrix}4/5\\3/5\end{pmatrix}}_{v_1},\underbrace{\begin{pmatrix}-3/5\\4/5\end{pmatrix}}_{v_2}\right\}$$

$$\forall\, v=\begin{pmatrix}x\\y\end{pmatrix}\in\mathbb{R}^2 \Rightarrow \exists!\, x',y'\in\mathbb{R}$$

$$\text{a.i.}\quad \underbrace{v = x'\cdot v_1 + y'\cdot v_2}$$

$$[v]_B=\begin{pmatrix}x'\\y'\end{pmatrix}$$

$$\begin{cases} x = 4/5\,x' - 3/5\,y' \\ y = 3/5\,x' + 4/5\,y' \end{cases}$$



$$\Rightarrow 27x'^2 + 2y'^2 = 50$$

$$\frac{x'^2}{\frac{\sqrt{50}}{27}} + \frac{y'^2}{\frac{\sqrt{50}}{2}} = 1$$

$$v_1 = 1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n$$
$$v_2 = 0 \cdot v_1 + 1 \cdot v_2 + \dots + 0 \cdot v_n$$

$$P = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \\ 0 & 0 & \dots & 1 \end{pmatrix} = I_n$$

$$B \xrightarrow{P} B'$$
$$B \xleftarrow{?\ P^{-1}} B'$$

$$[v]_B = P \cdot [v]_{B'}$$

Exemplu. a) $v = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ $\quad [v]_B = ?$

$B = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

b) $w \in \mathbb{R}^2 \quad [w]_B = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \quad w = ?$

c) $B' = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \quad P_B^{B'} = ?$

$[v]_{B'} = ? \quad [w]_{B'} = ?$

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a) $[v]_{B_c} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} = P_{B_c}^{B} \, [v]_B$

$P_{B_c}^{B} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = (v_1 \mid v_2)$

$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 \cdot e_1 + 2 \cdot e_2$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot e_1 + 1 \cdot e_2$

$\begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} [v]_B$

$$\Rightarrow [v]_B = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 \\ -1 \end{pmatrix} = (-1) \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ -1 \end{pmatrix} = (-4)\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 7\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \checkmark$$

b) $[w]_B = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

$$w = (-2)v_1 + 3 \cdot v_2$$

$$= (-2)\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

